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Aman Jyoti, Maninder Singh, V. K. Banga, Jatinder Kaur, J. Gowrishankar, Sarbeswara Hota, S. Murugan, A. Bhowmik & N. Ashok

Comparative Performance Evaluation of Machine Learning Classifiers for Multi-Class Intrusion Detection on the NSL-KDD Dataset

Aman Jyoti[1*], Maninder Singh[2], V.K. Banga[3], Jatinder Kaur[4], Gowrishankar J[5], Sarbeswara Hota[6], S. Murugan[7,] A Bhowmik[8,9], N. Ashok[10,*]

[1]Research Scholar (Ph.D.), Rayat Bahra University, Mohali, Punjab, India; ishasareen1@gmail.com
[2]Professor, Rayat Bahra University, Mohali, Punjab, India; director-research@rayatbahrauniversity.edu.in
[3]Professor, G.B. Pant Institute of Engineering and Technology, Pauri Garhwal, Uttarakhand, India; vijaykumar.banga@gmail.com
[3]Advanced Centre of Research and Innovation, School of Engineering and Technology, CGC University, Mohali – 140307, Mohali, Punjab, India; jatinder.cgctc@gmail.com
[5]Department of Computer Science Engineering, School of Engineering and Technology, JAIN (Deemed to be University), Bangalore, Karnataka, India; Email ID: gowrishankar.j@jainuniversity.ac.in
[6]Department of Computer Application, Siksha 'O' Anusandhan (Deemed to be University), Bhubaneswar, Odisha-751030, India; Email: sarbeswarahota@soa.ac.in
[7]Department of Computer Science and Engineering, Sathyabama Institute of Science and Technology, Chennai, Tamil Nadu, India; Email Id - murugan.cse@sathyabama.ac.in
[8]Department of Additive Manufacturing, Mechanical Engineering, SIMATS, Saveetha Institute of Medical and Technical Sciences, Thandalam, Chennai-602105, India; abhowmiktit@gmail.com
[9]Division of Research and Development, Lovely Professional University, Punjab, Phagwara, India
[10]Faculty of Mechanical Engineering, Jimma Institute of Technology, Jimma University, Jimma, Ethiopia
*Corresponding: ashok.nagaraj@ju.edu.et

Abstract

Intrusion Detection Systems (IDSs) play a vital role in safeguarding modern network infrastructures against increasingly sophisticated cyber threats. However, the high dimensionality of network traffic data and the presence of imbalanced attack classes often limit the effectiveness of conventional Machine Learning (ML) approaches. This study proposes a feature-driven Intrusion Detection (ID) framework that combines XGBoost-based feature selection with multiple ML classifiers to improve attack detection performance while reducing computational complexity. The NSL-KDD dataset is utilized to evaluate the proposed approach across five traffic classes: Benign, Denial of Service (DoS), Probe, Remote-to-Local (R2L), and User-to-Root (U2R). XGBoost feature ranking is employed to identify thirteen highly relevant features for each attack category, thereby reducing data dimensionality and eliminating redundant attributes. The selected features are subsequently evaluated using six ML classifiers, namely LightGBM, Voting Classifier, CatBoost, Multi-Layer Perceptron (MLP), AdaBoost, and Stochastic Gradient Descent (SGD). Performance assessment is conducted using Precision, Recall, F1-Score, confusion matrices, and cross-validation analysis. Experimental results demonstrate that ensemble-based models, particularly CatBoost and LightGBM, achieve superior performance for majority attack classes such as DoS and Probe, while all classifiers exhibit challenges in detecting minority classes such as R2L and U2R due to severe class imbalance. Cross-validation results confirm the robustness and stability of the selected feature subsets across different attack categories. Furthermore, a computational complexity analysis highlights the suitability of the proposed framework for practical and resource-constrained ID environments. The findings emphasize the effectiveness of feature optimization in enhancing classification performance and provide valuable insights for the development of efficient and scalable IDS solutions.

Keywords: Cross-validation, Intrusion Detection System, Machine Learning, XGBoost, Feature Selection, NSL-KDD.

I. INTRODUCTION

Currently, the primary concern for all businesses and institutions is to provide superior security to their respective networks. The creation of the IDS solves the limitations of firewalls and remote access monitoring in providing appropriate security against attacks. Despite the best efforts to secure the information, intruders succeed in obtaining critical information [1]. IDS provides a solution for such destructive operations. IDS works by detecting attacks before, during, and after they occur. The IDS monitors and analyzes both system and user activity, investigates system configuration and flaws, evaluates system and file dependability, identifies attack trends, and monitors policy violations at the user level [2, 3]. The growing complexity and frequency of cyber-attacks highlight the critical need for robust and intelligent IDS. Traditional signature-based detection methods frequently fail to identify novel and complex threats, making ML an appealing alternative due to its capacity to learn from patterns and respond to changing attack strategies [4]. Specifically, ML classifiers have shown great potential in spotting anomalies in network traffic and discriminating between regular and malicious activity [5, 6]. However, the efficiency of these classifiers might vary greatly based on the category of attacks, the dataset utilized, and the assessment criteria used.

The selection of the NSL-KDD dataset in the conducted study in the research paper is motivated by its standardized structure and widespread adoption in the literature, enabling consistent comparison with prior work. Additionally, its multi-class attack categorization makes it suitable for analyzing classifier behavior under imbalanced conditions.

Unlike studies that propose new models, this work focuses on a systematic and controlled comparative evaluation of widely used classifiers under identical experimental conditions. The primary objective is to analyze classifier behavior in the presence of severe class imbalance and to highlight their limitations in detecting minority attack categories, which remain a critical challenge in ID research. This paper examines and compares the effectiveness of many popular ML classifiers—LightGBM, Voting Classifier, CatBoost, MLP, AdaBoost, and SGD—in the domain of IDS. These models were chosen due to their different learning approaches, which ranged from ensemble methods and neural networks to linear optimization strategies. Each classifier is tested on the NSL-KDD dataset, which is a sophisticated and extensively acknowledged standard for testing IDS. The goal of the conducted study is to see how well these models perform in recognizing five types of network traffic: Benign, DoS, Probe, R2L, and U2R. Rather than focusing just on total RE, the evaluation takes into account numerous performance indicators, such as PR, RE, and F1, providing a more thorough picture of each model's strengths and shortcomings, particularly when dealing with imbalanced and minority classes. This research intends to contribute to the creation of more precise and adaptive IDS frameworks capable of better defending against a wide range of cyber threats by determining the most effective classifier(s) for various types of attacks.

The conducted study does not aim to propose a new ML model; instead, it focuses on providing a systematic and controlled evaluation of widely used classifiers to understand their strengths and limitations, particularly in the context of multi-class imbalance. Such diagnostic analysis is essential for identifying gaps in current Intrusion Detection (ID) approaches and guiding future research directions.

II. LITERATURE REVIEW

This section gives a complete review of the extant literature on the research subject. Key studies, techniques, and findings are thoroughly analyzed in Table 1 to identify current trends and gaps. The review provides the theoretical underpinning and contextual information for the current study.

Table 1. Literature Review

Author(s)	Techniques Used	Dataset(s)	Key Contributions	Findings / Results
Bayu Adhi Tama et al. (2017) [7]	Ensemble classifiers (Bagging, Boosting, Voting, Stacking, Rotation Forest); CNN, C4.5, SVM	Dataset not explicitly mentioned in the study	Compared ensemble methods for IDS tasks using different base classifiers	Boosting and Stacking performed best; Stacking gave the highest PR, Accuracy (ACC), AUC, and RE
Md. Raihan-Al-Masud et al. (2019) [8]	Ensemble ML with voting classifier	NSL-KDD	Proposed NIDS using ensemble ML to overcome limitations of traditional ML	Outperformed traditional ML; improved detection rate
Gu et al., 2019 [9]	DT-ENSVM (Ensemble SVM)	NSL-KDD	Developed a 3-layer ensemble for ID	99.41% ACC using FCM clustering
Zhang et al., 2019 [10]	FCN, VAE, Seq2Seq	Multiple public datasets	Evaluated deep learning models for anomaly detection	DL methods compared on ACC, PR, RE
Gao et al., 2019 [11]	Adaptive ensemble voting (DT, SVM, DNN)	NSL-KDD	Improved performance using adaptive classifier weights	85.2% ACC, 86.5% PR, 84.9% F1

Murk Marvi et al. (2020) [12]	Integrated Feature Selection (IFS); LightGBM	Dataset not explicitly mentioned in the study	Proposed a 3-stage IFS method to enhance DDoS detection	20% improvement in metrics; 77% feature reduction led to better performance
Yang et al., 2020 [13]	Data resampling strategy	NSL-KDD	Improved rare attack detection via instance multiplication	R2L/U2R DR increased significantly with more data
Huang & Lei, 2020 [14]	IGAN	NSL-KDD, CICIDS2017, UNSW-NB15	Solved class imbalance by generating minority class samples	IGAN outperformed 15 ML/DL methods
Jiang et al., 2020 [15]	CNN, Bi-LSTM, OSS, SMOTE	NSL-KDD, UNSW-NB15	NIDS combines spatial-temporal feature extraction	Achieved 83.58% (NSL-KDD), 77.16% (UNSW-NB15) ACC
Kasongo & Sun, 2020 [16]	ML with XGBoost for feature selection	UNSW-NB15	Improved IDS ACC using reduced feature space	Recommends NSL-KDD for further validation
Sun et al., 2020 [17]	CNN + LSTM hybrid (DL-IDS)	CICIDS2017	Spatial + temporal feature fusion with category weights	Achieved 98.67% ACC, 93.32% F1 score
Saranya et al., 2020 [18]	LDA, CART, RF	KDD-CUP	Compared classifiers for diverse network environments	RF achieved 99.65% ACC
Thakkar & Lohiya, 2020 [19]	Review study	Multiple datasets	Comparative dataset evaluation for IDS research	CICIDS2017 & CSE-CIC-IDS2018 most relevant
Andresini et al., 2020 [20]	MINDFUL (Autoencoders + 1D CNN)	KDD99, UNSWNB15, CICIDS2017	Multichannel deep feature learning for IDS	Up to 97.9% ACC (CICIDS2017)
Zhou et al., 2020 [21]	CFS-BA Ensemble (C4.5, RF, Forest PA)	NSL-KDD, CICIDS2017	Multi-attack classifier with feature correlation selection	Achieved 99% ACC; lacks time efficiency analysis
Karatas et al., 2020 [22]	kNN, RF, DT, GB, LDA with SMOTE	CSE-CIC-IDS2018	Balanced class distribution via SMOTE	ACC improved across ML models

III. RESEARCH GAP

The existing body of research on IDS using ML has made significant progress; however, several methodological and evaluation-related limitations remain inadequately addressed. A large proportion of prior studies focus primarily on binary classification, which fails to capture the complexity of real-world network traffic involving multiple attack categories. Although some works consider multi-class classification, their experimental setups often vary significantly in terms of preprocessing techniques, feature engineering, and validation strategies, leading to inconsistent and non-comparable results.

A critical gap lies not only in the presence of class imbalance in benchmark datasets such as NSL-KDD, but also in the lack of systematic analysis of how standard machine learning classifiers inherently behave under such imbalanced multi-class conditions. While several studies attempt to improve minority class detection using resampling or hybrid models, they often introduce additional algorithmic modifications, making it difficult to isolate and understand the baseline behavior and limitations of commonly used classifiers.

Furthermore, many existing works emphasize overall accuracy as the primary evaluation metric, which can be misleading in imbalanced scenarios. There is limited focus on class-wise performance evaluation, particularly for rare but critical attack categories such as R2L and U2R. As a result, the true weaknesses of widely used classifiers remain underexplored, especially when evaluated under uniform and controlled experimental settings. Although numerous intrusion detection studies have employed ML and ensemble learning techniques, many existing approaches primarily focus on achieving high overall accuracy while overlooking the detection performance of minority attack categories such as U2R and R2L attacks. Furthermore, several studies rely on a large number of input features, resulting in increased computational complexity, longer training times, and reduced suitability for real-time deployment. While feature selection methods have been explored in the literature, limited research has systematically investigated the impact of selecting a compact and optimized feature subset for individual attack categories using advanced feature-ranking techniques. Therefore, there remains a need for a framework that can reduce feature dimensionality while maintaining effective intrusion detection performance across multiple attack classes.

Another important limitation is the absence of standardized benchmarking frameworks where multiple classifiers are evaluated under identical preprocessing, feature representation, and training conditions. This lack of consistency prevents fair comparison and obscures the relative strengths and weaknesses of different learning paradigms.

In this context, the present study addresses a diagnostic research gap by conducting a systematic and controlled comparative evaluation of diverse machine learning classifiers on a multi-class intrusion detection problem using the NSL-KDD dataset. Unlike approaches that propose new models or apply data balancing techniques, this study deliberately evaluates classifiers under the dataset's natural imbalanced distribution to analyze their intrinsic behavior, bias toward majority classes, and limitations in detecting minority attacks. This provides a clearer and more realistic understanding of classifier performance, thereby establishing a reliable baseline for future research focused on imbalance mitigation and model enhancement.

IV. ATTACKS IN NSL-KDD

This section gives a detailed summary of the various kinds of attacks identified in the NSL-KDD dataset. The complete NSL-KDD dataset was utilized in this study. The dataset was divided into training and testing subsets following standard evaluation practices to ensure fair comparison. Table 2 displays each attack category according to its description and network security objectives. Representative examples are provided to highlight the variety and complexity of attack pathways. This analysis forms the basis for evaluating ID models in subsequent sections.

Table 2. Types of attacks in NSL-KDD

Attack Type	Description	Objective	Examples in NSL-KDD
Benign (Normal)	Identifies valid and non-malicious user behaviour [23]	Routine network operations	normal
Denial of Service (DoS)	Overwhelms network or system resources, preventing valid access [24]	Disrupts system availability	smurf, neptune, back, teardrop
Remote to Local (R2L)	Attempts to obtain illegal local access from a faraway place [25, 26]	Breach the system remotely	guess_passwd, ftp_write, warezclient
User to Root (U2R)	Attempts to escalate from user-level to root/administrator privileges [27]	Gain control of the system	buffer_overflow, perl, loadmodule
Probing (Surveillance)	Scans or probes the network to acquire information or detect vulnerabilities [28]	Network reconnaissance or mapping	nmap, satan, portsweep, ipsweep

V. SIGNIFICANCE OF LEARNING CLASSIFIERS IN IDS

Learning classifiers play a significant part in the fabrication of intelligent and efficient IDS. With the increasing incidence and sophistication of cyber-attacks, classic signature-based IDS models frequently fail to detect innovative or sophisticated incursions [29, 30]. Classifiers, which use ML and statistical models, allow IDS to effortlessly acquire patterns from historical network traffic and discriminate between normal and malicious activity [31]. This adaptability enables the successful detection of zero-day assaults in the absence of predefined criteria. Learning classifiers additionally decreases manual involvement and false positives, improving the detection process's reliability [32]. Techniques like decision trees, support vector machines, neural networks, and ensemble models help to achieve high ACC and robust performance. They provide binary and multiclass classification, permitting IDS to classify a variety of threats such as DDoS, probing, and ransomware. Classifiers are capable of handling high-dimensional, large-scale data, allowing for continuous evaluation in dynamic network contexts [33]. Additionally, they facilitate continuous learning, which makes IDS quicker to adapt to changing threat landscapes. Incorporating feature selection methods improves their PR by concentrating on the most important network characteristics [34]. Classifiers also enable hybrid detection procedures, which combine anomaly and misuse detection approaches. Their performance, particularly with optimized models such as LightGBM and SGD, makes them appropriate for current corporate networks. Overall, learning classifiers improve the understanding, speed, and precision of IDS, making them an essential aspect of cybersecurity systems. Fig. 1 showcases the importance and impact of learning algorithms in strengthening IDS capabilities.

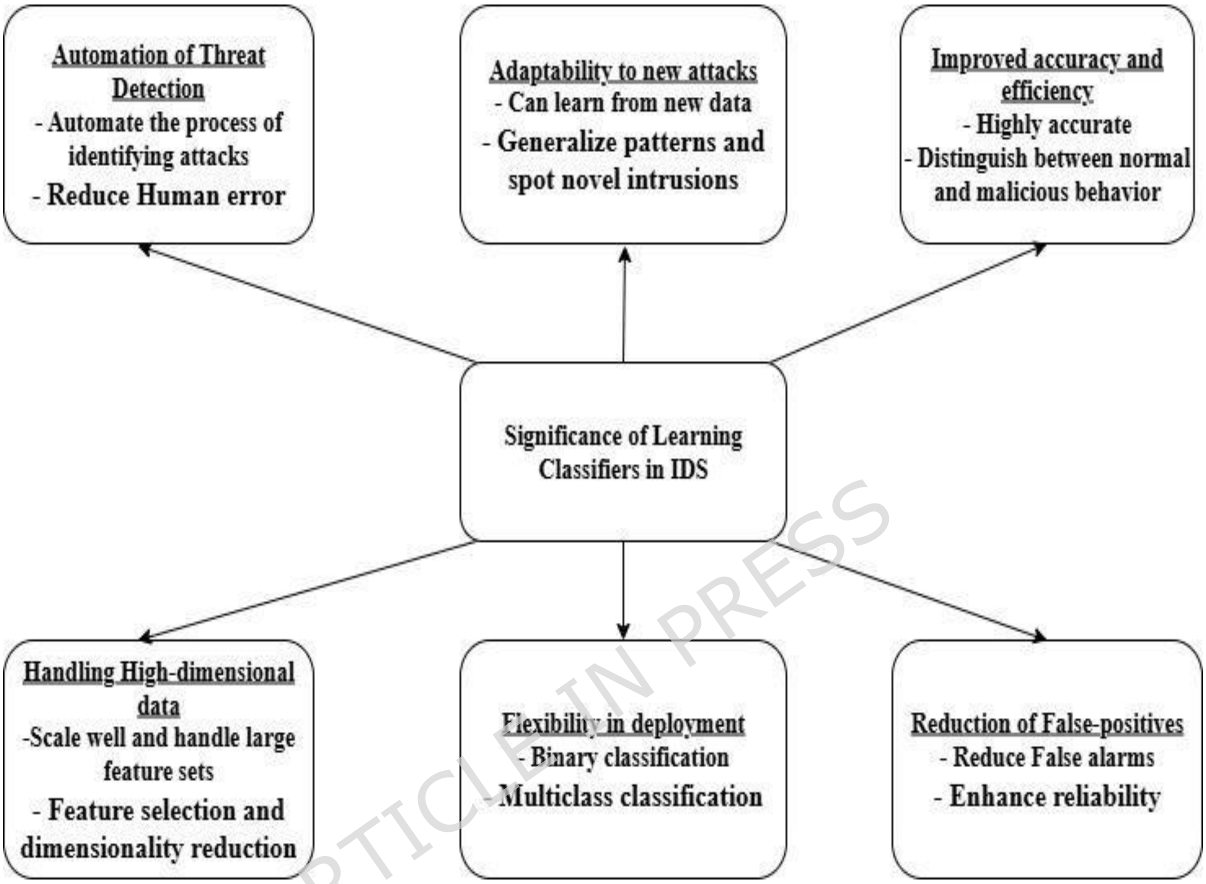

Fig. 1 Significance of learning classifiers in IDS

The selected classifiers represent diverse learning paradigms, including ensemble learning (Voting, AdaBoost, LightGBM), gradient boosting (CatBoost), neural networks (MLP), and linear models (SGD). This selection enables a comprehensive comparison of different algorithmic approaches under a unified experimental framework, addressing gaps identified in prior studies.

Table 3 provides an overview of five ML classifiers relevant to the study. Each classifier is briefly described, including its essential concept and underlying operating mechanism. The operational principles focus on how these models process and learn from data. The table also describes the primary advantages of every approach in practical usage. Potential limits or disadvantages are also addressed in order to provide a balanced review.

Table 3. Overview of ML classifiers

Classifier	Brief Introduction	Working Principle	Advantages	Disadvantages
LightGBM	A histogram-based gradient boosting framework for large datasets [35]	Grows trees leaf-wise, uses GOSS and EFB to reduce data and variables	- Fast training - Low memory use - High RE - Handles large datasets - GPU support	- Risk of overfitting on small datasets - Limited documentation/user base
Voting Classifier	An ensemble method that combines predictions of multiple models [36]	Combines models using hard voting (majority) or soft voting (average probabilities)	- Reduces individual model bias - Works with diverse models - Improves general RE	- Needs diverse base models - May be complex to interpret

CatBoost	Gradient boosting algorithm optimized for categorical features [36, 37]	Uses statistical encoding of categorical variables; works well without preprocessing	- Handles categorical/text/audio/image- Less overfitting - High RE - Supports Python, R, and GPU	- Minor difficulty in parameter tuning for categorical data
MLP (Multi-layer Perceptron)	A fully connected feedforward neural network model [38]	Uses multiple hidden layers for learning from data via backpropagation	- Learns directly from data - No assumptions on data distribution - Capable of modeling complex patterns	- Inefficient with large parameters - Requires long training time - Poor with sequential data
AdaBoost	An ensemble learning method that combines multiple weak learners to form a strong classifier	Sequentially trains weak models, giving higher weight to misclassified samples to improve performance	-Improves RE -Reduces bias -Works well with weak learners	-Sensitive to noise -Prone to overfitting with outliers -Requires careful parameter tuning
SGD Classifier	An optimization-based classifier using stochastic gradient descent [39, 40]	Updates weights using one random data point per iteration; suited for large data	- Fast for large datasets - Low memory usage - Escapes local minima - Best for convex problems	- Noisy convergence - Hard to optimize with many parameters Poor vectorization support

VI. NOVEL CONTRIBUTION

The present study introduces a feature-driven ID framework that combines XGBoost-based feature ranking with multiple ML classifiers for comprehensive attack detection. Unlike conventional approaches that utilize the entire feature space, the proposed framework identifies the most influential features separately for DoS, Probe, R2L, and U2R attack categories and evaluates their effectiveness using six different classifiers, namely LightGBM, Voting Classifier, CatBoost, MLP, AdaBoost, and SGD. The study provides a detailed comparative analysis based on precision, recall, F1-score, confusion matrices, and cross-validation results. This approach offers valuable insights into the relationship between feature relevance and classifier performance, thereby contributing to the development of computationally efficient IDS.

VII. METHODOLOGY AND EXPERIMENTAL SETUP

The experimental methodology adopted in this study is designed to provide a systematic, controlled, and reproducible evaluation framework for analyzing the behavior of multiple ML classifiers on a multi-class ID problem. The primary objective is to ensure fair comparison across models under identical conditions while enabling a detailed assessment of their strengths and limitations, particularly in the presence of class imbalance.

The experiments are conducted using the NSL-KDD dataset, which contains both numerical and categorical features representing diverse network traffic characteristics. Before model training, the dataset undergoes a structured preprocessing pipeline. Categorical attributes such *as* protocol type, service, and flag are transformed into numerical representations using one-hot encoding to ensure compatibility with ML algorithms. To maintain uniformity in feature contribution and prevent scale dominance, numerical features are normalized using the StandardScaler() Method. The dataset is examined for inconsistencies, redundancies, and missing values; however, as NSL-KDD is a preprocessed benchmark dataset, no missing entries are observed. This preprocessing ensures that the input data is consistent, normalized, and suitable for comparative analysis.

To ensure reproducibility and consistency, the dataset is divided into training and testing subsets using an 80:20 split with a fixed random seed. This controlled split ensures that all classifiers are evaluated on the same data distribution, eliminating variability arising from different sampling strategies. Unlike optimization-focused studies, this work emphasizes comparative behavioral analysis and therefore maintains a consistent evaluation protocol across all models.

A diverse set of ML classifiers—LightGBM, CatBoost, Voting Classifier, MLP, AdaBoost, and Stochastic Gradient Descent (SGD)—is selected to represent different learning paradigms, including ensemble learning, gradient boosting, neural networks, and linear models. This selection enables a comprehensive evaluation of how different algorithmic approaches respond to the same ID problem. To ensure fairness in comparison, all classifiers are initially implemented using their default hyperparameters. Subsequently, selected models such as LightGBM and CatBoost are further optimized using grid search techniques to evaluate the impact of tuning on performance. The tuning process is applied consistently and transparently, ensuring that improvements are attributable to model capability rather than experimental bias.

Feature selection techniques, including Recursive Feature Elimination (RFE) and XGBoost-based ranking, were incorporated to identify the most relevant features for different attack categories.

A key characteristic of the NSL-KDD dataset is its highly imbalanced class distribution, where majority classes such as Benign and DoS dominate, while minority classes such as R2L and U2R are significantly underrepresented. Instead of applying resampling or cost-sensitive techniques, this study intentionally evaluates classifiers under the dataset's natural imbalance conditions. This decision is made to analyze the intrinsic learning behavior and bias of classifiers when exposed to real-world skewed data distributions. By avoiding external balancing interventions, the study provides a clearer understanding of how different models inherently respond to minority classes. The impact of class imbalance is thoroughly examined in the results section, and its implications for intrusion detection performance are critically discussed.

To ensure a comprehensive assessment, classifier performance is evaluated using class-wise Precision, Recall, and F1-score. These metrics are particularly suitable for imbalanced datasets, as they provide a more informative measure of model effectiveness than overall accuracy. The evaluation focuses on both aggregate performance and class-specific behavior, enabling detailed analysis of model strengths and weaknesses across different attack categories.

The proposed framework differs from existing ID studies in several important aspects. First, rather than applying a common feature set across all attack categories, it performs attack-specific feature selection using XGBoost ranking, allowing the framework to capture the unique characteristics associated with different intrusion types. Second, the study evaluates a diverse set of machine learning and ensemble learning classifiers within a unified experimental environment, enabling a fair and comprehensive performance comparison. Third, the framework emphasizes both feature optimization and classifier evaluation, providing insights not only into detection accuracy but also into model robustness and generalization capability. Unlike many previous studies that focus primarily on maximizing accuracy, the proposed approach investigates the behavior of classifiers across majority and minority attack classes, thereby offering a more realistic assessment of intrusion detection effectiveness in practical network security scenarios.

All experiments are conducted under a unified and controlled framework, including identical preprocessing steps, data splits, and evaluation criteria. The use of a fixed random seed, consistent feature transformations, and standardized metric computation ensures that the results are reproducible and comparable. This controlled setup enables a reliable assessment of classifier behavior without the influence of external variability.

A. Data Preprocessing and Feature Engineering

This section elaborates the flowchart in Fig. 2 for the procedure adopted to implement the feature selection in IDS, followed by the results achieved. The *KDDTrain+_2.csv* and *KDDTest+_2.csv* have been used for training and testing.

Feature reduction plays a crucial role in improving the efficiency and practicality of IDS. In this study, the number of selected features is restricted to thirteen for each attack category to achieve an optimal balance between detection capability and computational overhead. The selection of a compact feature subset helps eliminate redundant and less informative attributes, reduces the dimensionality of the dataset, accelerates model training and testing processes, and minimizes the risk of overfitting. By concentrating on the most discriminative features, the framework enables classifiers to focus on the most relevant attack characteristics while maintaining competitive detection performance. This feature optimization strategy is particularly beneficial for resource-constrained and real-time cybersecurity environments.

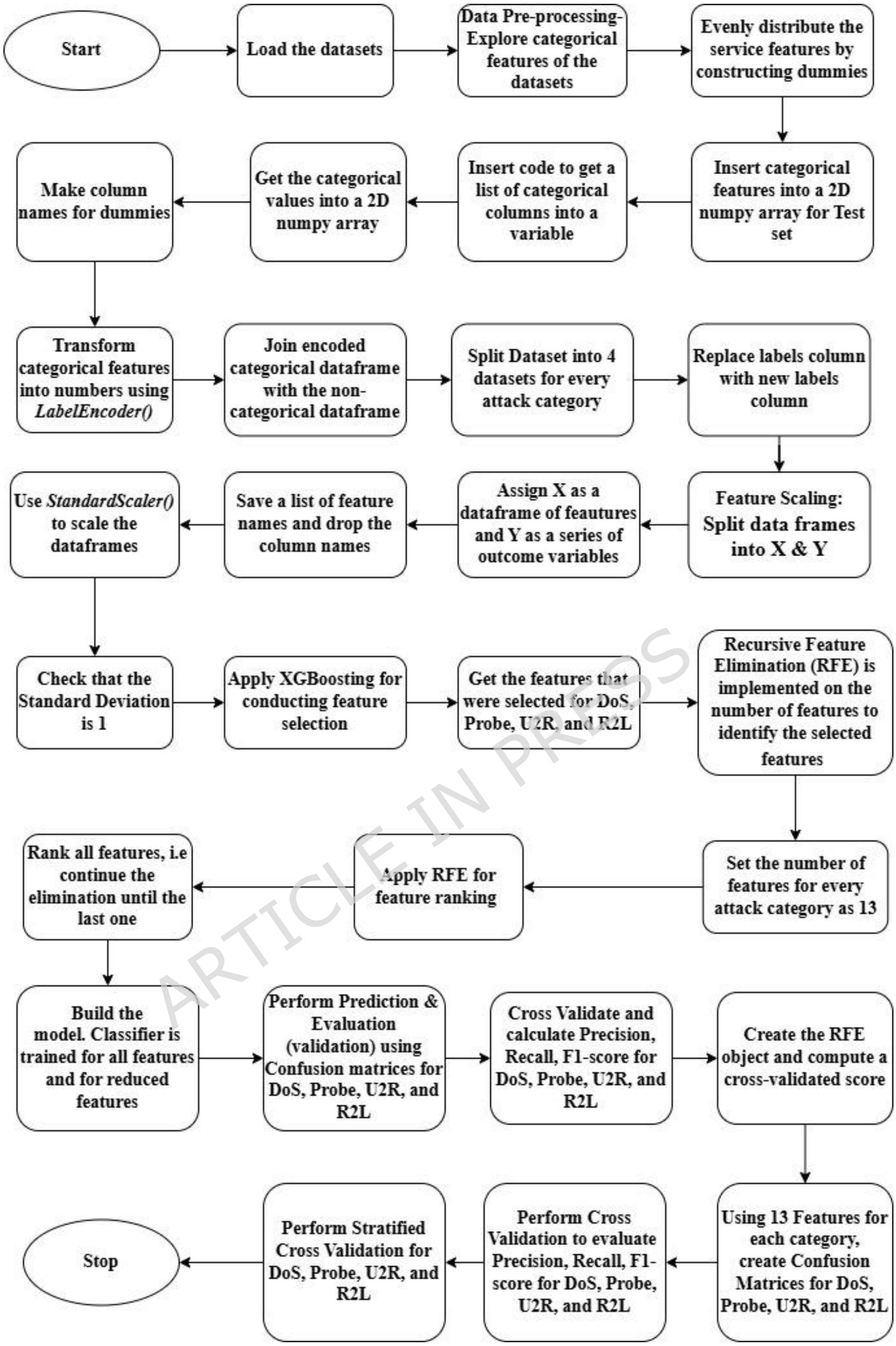

Fig. 2 Methodology for Feature Engineering, Selection, and Classification of Network Intrusion Attack Categories

Fig. 2 illustrates the complete methodological workflow adopted for preprocessing, feature engineering, feature selection, and classification of network intrusion attack categories. The process begins with loading the dataset and performing data preprocessing by exploring categorical attributes and converting them into numerical representations through dummy variable construction and label encoding techniques. The encoded categorical features are then merged with non-categorical attributes to form a unified DataFrame. Subsequently, the dataset is divided according to different attack categories, namely DoS, Probe, U2R, and R2L, followed by replacement of label columns and separation of independent variables (X) and dependent variables (Y). Feature scaling is performed using the StandardScaler() method to normalize the data and ensure a standard deviation of one. Recursive Feature Elimination (RFE) is then applied to identify the most significant features for each attack category, while XGBoosting is utilized for feature selection and ranking. The methodology further constrains the number of selected features to thirteen for each attack category.

Thereafter, a Decision Tree classifier is constructed and trained using both the complete and reduced feature sets.

Finally, the model performance is evaluated through prediction, confusion matrices, and stratified cross-validation using performance metrics such as Precision, Recall, and F1-score for DoS, Probe, U2R, and R2L attack classes.

B. Dataset Distribution and Feature Selection

The results obtained after the successful implementation of the procedure adopted in the flowchart in Fig. 2 and the algorithm are mentioned below.

The dataset was split into training and testing sets using an 80:20 ratio. To ensure reproducibility, a fixed random seed was used. Table 4 demonstrates the name of the attacks and the number in which they exist in the NSL KDD dataset. The normal (non–suspicious) is 67343, which is the maximum number, and Spy is 2, in minimum number. Table 4 demonstrates the label distribution for the training set.

Table 4. Label distribution for Training Set and Testing Set

Training Set		**Testing Set**	
Attacks	**In Number**	**Attacks**	**In Number**
Normal	67343	Normal	9711
Neptune	41214	Neptune	4657
Satan	3633	Guess_passwd	1231
Ipsweep	3599	Mscan	996
Portsweep	2931	Warezmaster	944
Smurf	2646	Apache2	737
Nmap	1493	Satan	735
Back	956	Processtable	685
Teardrop	892	Smurf	665
Warezclient	890	Back	359
Pod	201	Snmpguess	331
Guess_passwd	53	Saint	319
Buffer_overflow	30	Mailbomb	293
Warezmaster	20	Snmpguess	331
Land	18	Saint	319
Imap	11	Mailbomb	293
Rootkit	10	Snmpgetattack	178
Loadmodule	9	Portsweep	157
ftp_write	8	Ipsweep	141
Multihop	7	Httptunnel	133
Phf	4	Nmap	73
Perl	3	Pod	41
Spy	2	Buffer_overflow	20
		Multihop	18
		Named	17
		Ps	15
		Sendmail	14
		Xterm	13
		Rootkit	13
		Teardrop	12
		Xlock	9
		Land	7
		Xsnoop	4
		ftp_write	3
		Udpstorm	2
		Sqlattack	2
		Phf	2
		Worm	2
		Loadmodule	2
		Perl	2
		Imap	1

Table 5 depicts the dimensions of different attacks in accordance with the Training Set and Testing set.

Table 5. Table depicts the dimensions of different attacks as per Training Set and Testing set

Training set			Testing set	
Attack	**Dimensions**		**Attack**	**Dimensions**
DoS	113270, 123		DoS	17171, 123
Probe	78999, 123		Probe	12132, 123
R2L	68338, 123		R2L	12596, 123
U2R	67395, 123		U2R	9778, 123

Table 6 presents the various features selected through the XGBoost feature selection technique for the four attack categories.

Table 6. Features Selected Using XGBoost for Four Attack Categories

Features selected (DoS)	Features selected (Probe)	Features selected (R2L)	Features selected (U2R)
Logged_in	Logged_in	Src_bytes	Urgent
Count	Rerror_rate	Dst_bytes	Hot
Serror_rate	Srv_rerror_rate	Hot	Root_shell
Srv_serror_rate	Dst_host_srv_count	Num_failed_logins	Num_file_creations
Same_srv_rate	Dst_host_diff_srv_rate	Is_guest_login	Num_shells
Dst_host_count	Dst_host_same_src_port_rate	Dst_host_srv_count	Srv_diff_host_rate
Dst_host_srv_count	Dst_host_srv_diff_host_rate	Dst_host_same_src_port_rate	Dst_host_count
Dst_host_same_srv_rate	Dst_host_rerror_rate	Dst_host_srv_diff_host_rate	Dst_host_srv_count
Dst_host_serror_rate	Dst_host_srv_rerror_rate	Service_ftp	Dst_host_same_src_port_rate
Dst_host_srv_serror_rate	Protocol_type_icmp	Service_ftp_data	Dst_host_srv_diff_host_rate
Service_http	Service_eco_i	Service_http	Service_ftp_data
Flag_S0	Service_private	Service_imap4	Service_http
Flag_SF	Flag_SF	Flag_RSTO	Service_telnet

Table 7 shows the ranking of Selected Features for Four Attack Categories Using the XGBoost Feature Selection Technique.

Table 7. Ranked Features Selected Using XGBoost for Four Attack Categories

Rank	Features selected (DoS)	Features selected (Probe)	Features selected (R2L)	Features selected (U2R)
1	Same_srv_rate	dst_host_same_src_port_rate	src_bytes	Hot
2	Count	dst_host_srv_count	dst_bytes	dst_host_srv_count
3	Flag_SF	dst_host_rerror_rate	Hot	dst_host_count
4	Dst_host_serror_rate	service_private	dst_host_srv_diff_host_rate	root_shell
5	Dst_home_same_srv_rate	logged_in	service_ftp_data	num_shells
6	Dst_host_srv_count	dst_host_diff_srv_rate	dst_host_same_src_port_rate	service_ftp_data
7	Dst_host_count	dst_host_srv_diff_host_rate	dst_host_srv_count	dst_host_srv_diff_host_rate

8	Logged_in	flag_SF	num_failed_logins	num_file_creations
9	Serror_rate	service_eco_i	service_imap4	dst_host_same_src_port_rate
10	Dst_host_srv_serror_rate	rerror_rate	is_guest_login	service_telnet
11	Srv_serror_rate	Protocol_type_icmp	service_ftp	srv_diff_host_rate
12	Service_http	dst_host_srv_rerror_rate	flag_RSTO	service_http
13	Flag_S0	srv_rerror_rate	flag_RSTO	urgent

VIII. RESULTS AND DISCUSSIONS

A. Classifier implementation and confusion matrices

LightGBM

This section presents an analysis of the confusion matrix derived from the application of the LightGBM classifier. The evaluation is conducted using the NSL-KDD dataset. Relevant performance tables are generated to interpret the classifier's predictive PR, RE, and F1 (F1S). Fig. 3 shows the confusion matrix of LightGBM.

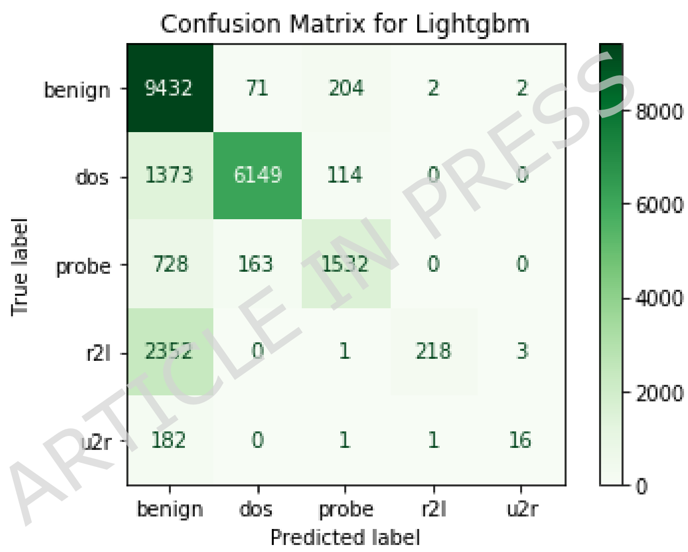

Fig. 3. Confusion matrix for LightGBM

Table 8 depicts the calculated values of performance evaluation metrics for LightGBM.

Table 8. Performance evaluation metrics (LightGBM)

Class	TP	FP	FN	PR	RE	F1S
Benign	9432	3635	279	72.2	97.1	82.9
DoS	6149	234	1487	96.3	80.6	87.6
Probe	1532	320	891	82.7	63.2	71.7
R2L	218	3	2356	98.6	8.5	15.4
U2R	16	21	184	43.2	8	13.5

LightGBM performs well in the Benign and DoS classes, demonstrating that it can recognize both normal and high-frequency attack traffic. The Benign class achieved 72.2% PR and 97.1% RE, yielding an F1S of 82.9%. This indicates that, while most normal traffic was appropriately categorized, there were a significant number of FPs. Similarly, the DoS class achieved a high PR of 96.3%, resulting in an F1 of 87.6%, making it the top-performing assault category overall. The Probe class performs moderately, with 82.7% PR and 63.2% RE. This resulted in an F1S of 71.7%, indicating that the model is rather good at detecting this type of probing activity; however, there is still potential for improvement due to the high FNs. The model operates poorly in the R2L and U2R classes, which reflect low-frequency, sophisticated attacks. R2L had only 8.5% RE despite a high PR of 98.6%, resulting in a very low F1S of 15.4%. This means that, while the model produced very few FPs, it failed to detect the vast majority of actual R2L instances. U2R detection is also poor, with only 43.2% PR, 8.0% RE, and an F1S of 13.5%, showing that this key attack type is mostly missed by the model.

Voting Classifier

This section analyzes the confusion matrix generated by the Voting classifier. Relevant performance tables are created to explain the classifier's prediction PR, RE, and F1S. Fig. 4 depicts the confusion matrix of the Voting classifier.

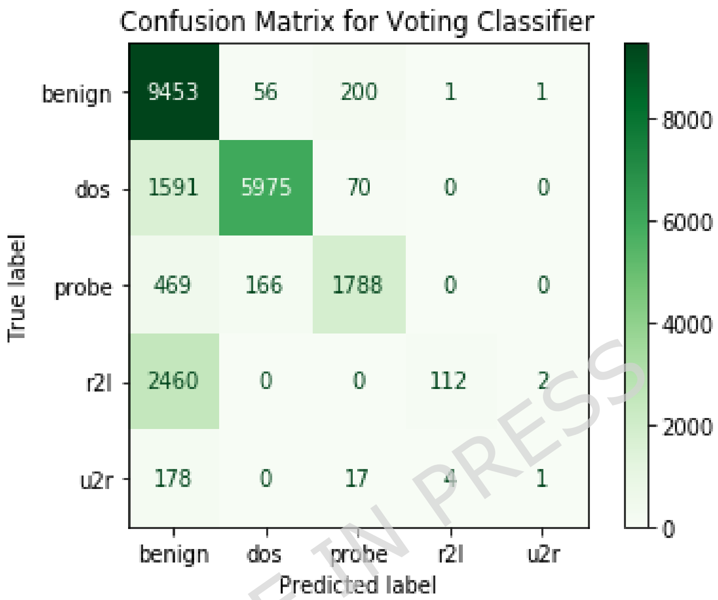

Fig. 4. Confusion matrix for Voting classifier

Table 9 depicts the calculated values of performance evaluation metrics for the Voting classifier.

Table 9. Performance evaluation metrics (Voting classifier)

Class	TP	FP	FN	PR	RE	F1S
Benign	9453	4298	258	68.7	97.3	80.6
DoS	5975	222	1661	96.4	78.2	86.3
Probe	1788	287	635	86.2	73.8	79.5
R2L	112	5	2462	95.7	4.3	8.1
U2R	1	3	199	25	0.5	0.9

The performance evaluation of the classification model across different classes—Benign, DoS, Probe, R2L, and U2R—reveals varying effectiveness in terms of PR, RE, and F1S. The Benign class shows strong performance, with high RE (97.3%), though PR is moderate (68.7%), leading to a balanced F1S of 80.6%. The DoS class maintains a high PR of 96.4% but shows a relatively lower RE (78.2%), resulting in an F1S of 86.3%. For the Probe class, PR and RE stand at 86.2% and 73.8%, respectively, yielding an F1S of 79.5%, indicating fair detection but room for improvement. In contrast, the model struggles significantly with the R2L and U2R classes; although R2L has a PR of 95.7%, its RE is extremely low at 4.3%, leading to an F1S of just 8.1%. The U2R class exhibits the weakest performance overall, with a RE of only 0.5%, a PR of 25%, and an F1S of 0.9%, indicating a substantial challenge in correctly identifying these rare attack types.

CatBoost

This section analyzes the confusion matrix generated by the CatBoost classifier. Relevant performance tables are created to explain the classifier's prediction PR, RE, and F1S. Fig. 5 depicts the confusion matrix of the CatBoost classifier.

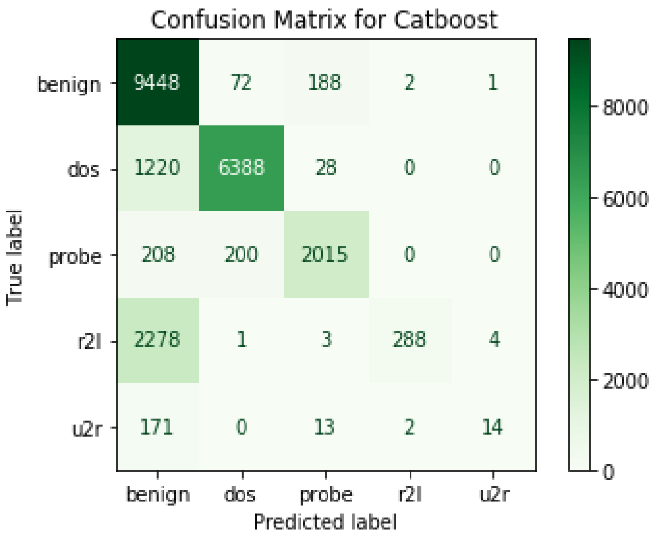

Fig. 5. Confusion matrix for CatBoost classifier

Table 10 depicts the calculated values of performance evaluation metrics for the CatBoost classifier.

Table 10. Performance evaluation metrics (CatBoost classifier)

Class	TP	FP	FN	PR	RE	F1S
Benign	9448	5877	263	61.7	97.3	75.6
DoS	6388	273	1248	95.9	83.6	89.3
Probe	2015	232	408	89.6	83.1	86.2
R2L	288	4	2286	98.6	11.2	20.2
U2R	14	5	186	73.7	7.0	12.9

The evaluation of classification performance across the five classes shows a mix of strengths and weaknesses in the model's ability to distinguish between different types of network traffic. For the Benign class, the model achieves high RE (97.3%), indicating strong detection of normal traffic; however, the low PR (61.7%) suggests a considerable number of false positives, lowering the F1S to 75.6%. The DoS class exhibits robust overall performance, with high PR (95.9%) and good RE (83.6%), resulting in a strong F1S of 89.3%, indicating the classifier is reliable in identifying DoS attacks. Similarly, the Probe class performs well, with balanced metrics—PR at 89.6% and RE at 83.1%, leading to an F1S of 86.2%—reflecting effective classification of these attacks. However, performance drops significantly for the R2L class; although PR is exceptionally high at 98.6%, the RE is extremely low (11.2%), causing a low F1S of 20.2%, revealing the model's tendency to miss most R2L instances. The U2R class shows the weakest performance overall, with RE at just 7.0% and an F1S of 12.9%, despite a moderately high PR of 73.7%. This suggests that the classifier struggles to detect rare and complex attack types like R2L and U2R, highlighting the need for improved sensitivity toward minority classes.

MLP (Multi-Layer Perceptron)

This section presents an analysis of the confusion matrix derived from the application of the MLP. Relevant performance tables are generated to interpret the classifier's predictive PR, RE, and F1S. Fig. 6 shows the confusion matrix of the MLP classifier.

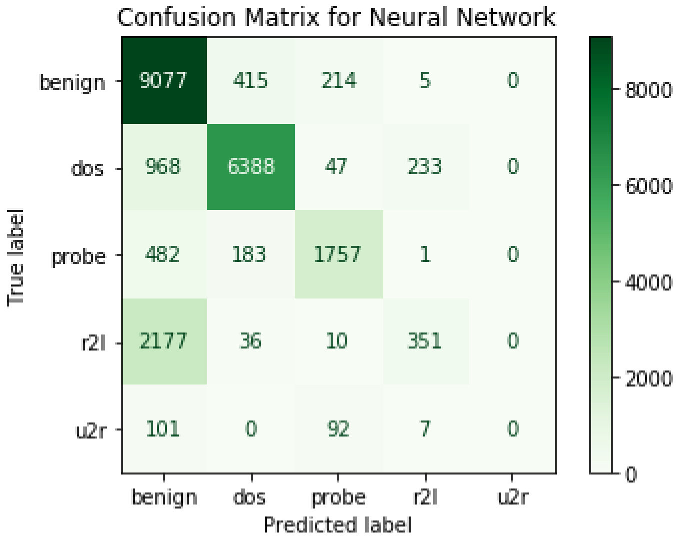

Fig. 6. Confusion matrix for MLP

Table 11 depicts the calculated values of performance evaluation metrics for the MLP classifier.

Table 11. Performance evaluation metrics (MLP)

Class	TP	FP	FN	PR	RE	F1S
Benign	9077	3728	634	70.9	93.5	80.7
DoS	6388	634	1248	90.9	83.6	87.1
Probe	1757	363	666	82.9	72.5	77.3
R2L	351	246	2223	58.8	13.6	22.1
U2R	0	0	200	—	0.0	0.0

The evaluation of the classifier's performance across the five NSL-KDD classes reveals varied effectiveness in terms of PR, RE, and F1S. For the Benign class, the model achieves a high RE of 93.5%, indicating it successfully identifies most normal traffic, though the PR is moderate at 70.9% due to a notable number of false positives. This results in a respectable F1S of 80.7%, reflecting overall balanced performance. The DoS class also performs well, with a strong PR of 90.9% and RE of 83.6%, leading to a high F1S of 87.1%, suggesting reliable detection of DoS attacks. For the Probe class, the model yields a reasonable PR (82.9%) and RE (72.5%), with an F1S of 77.3%, though there is room for improvement in identifying more subtle probing attacks. The performance sharply declines for the R2L class, where RE is very low (13.6%) and the F1S drops to 22.1%, despite a moderate PR of 58.8%, indicating the model frequently misses these attack types. The U2R class shows the poorest results—no true positives were detected, leading to zero PR, RE, and F1S, highlighting a complete failure to classify this rare and complex attack category. These results suggest the model performs well on the majority classes but struggles significantly with underrepresented and complex attack types, warranting further improvements in handling data imbalance and minority class detection.

AdaBoost

This section presents an analysis of the confusion matrix derived from the application of AdaBoost. The evaluation is conducted using the NSL-KDD dataset. Relevant performance tables are generated to interpret the classifier's predictive PR, RE, and F1S. Fig. 7 shows the confusion matrix of the AdaBoost classifier.

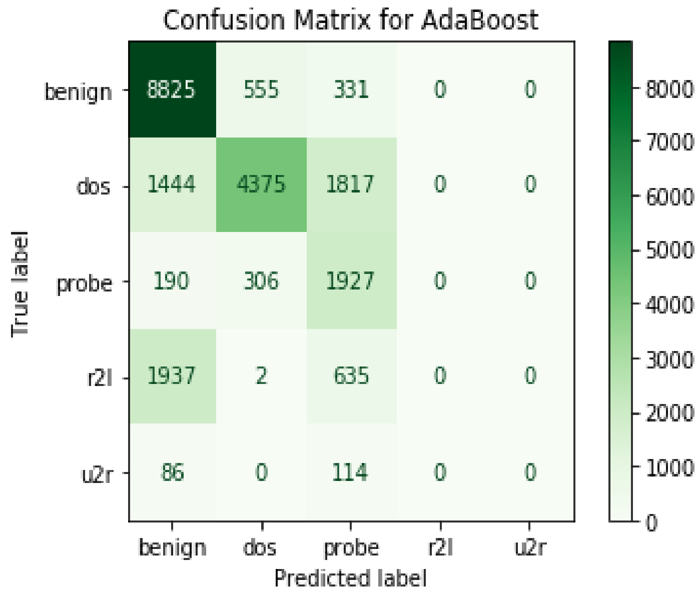

Fig. 7 Confusion matrix for AdaBoost

Table 12 depicts the calculated values of performance evaluation metrics for the AdaBoost classifier.

Table 12. Performance evaluation metrics (AdaBoost)

Class	TP	FP	FN	PR	RE	F1S
Benign	8825	5657	886	60.9	90.9	72.8
DoS	4375	863	3261	83.5	57.3	67.8
Probe	1927	2897	496	39.9	79.5	53.1
R2L	0	0	2574	—	0.000	0.000
U2R	0	0	200	—	0.000	0.000

The performance evaluation across the five classes reveals uneven classification results, highlighting both strengths and significant weaknesses in the model. The Benign class demonstrates high RE (90.9%), indicating the model effectively identifies most normal traffic. However, the PR is relatively low at 60.9%, suggesting a high number of false positives, which brings the F1S to a moderate 72.8%. For the DoS class, the model achieves strong PR (83.5%) but struggles with RE (57.3%), leading to an F1S of 67.8%. This indicates that while most predicted DoS instances are correct, many actual attacks are missed. The Probe class reflects an opposite pattern, with good RE (79.5%) but low PR (39.9%), indicating many false positives and resulting in an F1S of 53.1%. Performance for the R2L and U2R classes is critically poor; the model fails to identify any instances of these attacks, resulting in zero PR, RE, and F1S. This stark gap in performance underscores the model's inability to detect rare or low-frequency attack types, pointing to a severe class imbalance problem and the need for targeted improvements in detecting minority classes.

SGD

This section presents an analysis of the confusion matrix derived from the application of the SGD. The evaluation is conducted using the NSL-KDD dataset. Relevant performance tables are generated to interpret the classifier's predictive PR, RE, and F1S. Fig. 8 shows the confusion matrix of the SGD classifier.

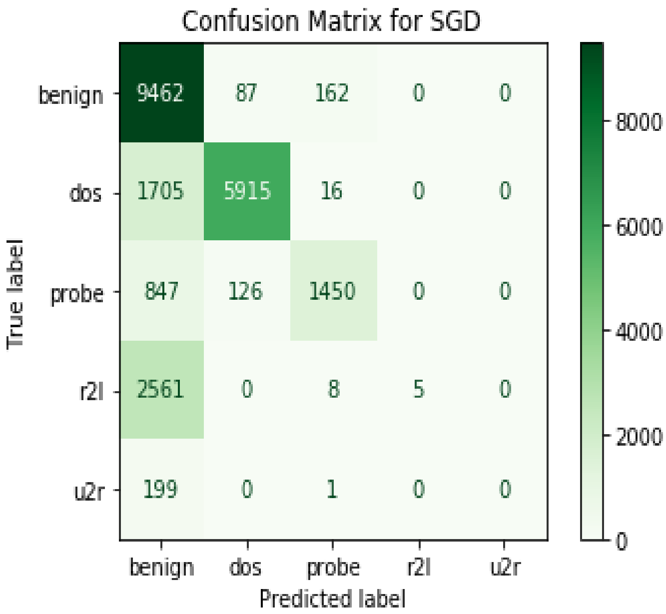

Fig. 8. Confusion matrix for SGD

Table 13 depicts the calculated values of performance evaluation metrics for the SGD classifier.

Table 13. Performance evaluation metrics (SGD)

Class	TP	FP	FN	PR	RE	F1S
Benign	9462	5312	249	64	97.4	77.2
DoS	5915	213	1721	96.5	77.5	85.9
Probe	1450	187	973	88.6	59.8	71.2
R2L	5	0	2569	1.0	2.0	4.0
U2R	0	0	200	----	0.0	0.0

The classification performance across the five classes shows significant variation, with strong results in the majority classes and poor detection of minority attack types. The Benign class achieves high RE (97.4%), indicating the model correctly identifies most normal traffic. However, the PR is moderate at 64%, due to a considerable number of false positives, resulting in an F1S of 77.2%, showing overall solid but improvable performance. The DoS class also performs strongly, with a high PR of 96.5% and a respectable RE of 77.5%, yielding an F1S of 85.9%, indicating effective identification of DoS attacks. The Probe class demonstrates decent PR at 88.6%, but a lower RE (59.8%) reduces its F1S to 71.2%, suggesting many probe attacks are missed despite correct predictions when made. In contrast, the model performs very poorly on the R2L class, achieving near-zero RE (2.0%) and a minimal F1S of 4.0%, despite perfect PR (1.0) for the few true positives it predicts. The U2R class exhibits the weakest performance, with no true positives detected at all, resulting in zero PR, RE, and F1S. Overall, while the model is effective in classifying the majority of traffic and attacks, its inability to detect rare classes like R2L and U2R highlights the need for better handling of class imbalance and improved sensitivity to low-frequency threats.

The extremely low recall observed for R2L and U2R classes across all classifiers indicates that standard ML models are inherently biased toward majority classes. This bias arises due to the disproportionate distribution of training samples, where minority classes fail to contribute significantly to model learning. As a result, models tend to misclassify rare attacks as normal or other dominant categories.

B. Cross-validation results

Cross-validation was applied in the current study; however, all models were evaluated under identical conditions to ensure fairness. Each classifier is evaluated using confusion matrices and standard performance metrics, including Precision, Recall, and F1-score, to assess its effectiveness in multi-class ID.

For the four attacks, DoS, Probe, U2R, and R2L, Table 14 shows the Cross-Validation readings of three performance evaluation parameters: Precision, Recall, and F1-Score.
It is important to note that the cross-validation results presented in Table 14 below are obtained using attack-specific datasets after feature selection and category-wise data preparation. These results evaluate the classification performance for individual attack categories under cross-validation settings and therefore are not directly comparable to the multi-class confusion matrix results presented in Tables 8–13. The confusion matrices evaluate the classifiers in a complete multi-class intrusion detection environment, whereas the cross-validation analysis assesses the stability and discriminative capability of the selected features within individual attack categories. Consequently, variations in performance metrics between the two evaluations are expected and reflect differences in the experimental objectives and data configurations.

Table 14. Cross-validation readings of performance evaluation parameters

Parameters	Attacks			
	DoS	Probe	R2L	U2R
Precision	0.99505 (+/- 0.00477)	0.99392 (+/- 0.00684)	0.97151 (+/- 0.01736)	0.86295 (+/- 0.08961)
Recall	0.99665 (+/- 0.00483)	0.99267 (+/- 0.00405)	0.96958 (+/- 0.01379)	0.90958 (+/- 0.09211)
F1-score	0.99585 (+/- 0.00392)	0.99329 (+/- 0.00512)	0.97051 (+/- 0.01478)	0.88210 (+/- 0.06559)

Table 14 evidences a high level of classification robustness across the majority of attack classes, with marginal degradation observed in more intricate categories. Specifically, the model achieves near-optimal performance for DoS and Probe attacks, with precision and recall consistently exceeding 0.99 and minimal variance, yielding F1-scores of 0.99585 (±0.00392) and 0.99329 (±0.00512), respectively—indicative of strong generalization and stability. In contrast, R2L attacks exhibit a moderate reduction in performance, with precision and recall values of 0.97151 (±0.01736) and 0.96958 (±0.01379), resulting in an F1-score of 0.97051 (±0.01478). This suggests increased sensitivity to feature ambiguity and intra-class variability. Performance further declines for U2R attacks, where precision (0.86295 ± 0.08961) and recall (0.90958 ± 0.09211) demonstrate higher dispersion, producing an F1-score of 0.88210 (±0.06559). The elevated standard deviations reflect instability and limited discriminative capability, likely attributable to class imbalance and the inherent sparsity and complexity of U2R patterns.

Overall, the model demonstrates excellent efficacy for high-frequency attack classes (DoS, Probe), reasonable generalization for moderately complex classes (R2L), and comparatively constrained yet acceptable performance for low-frequency, high-complexity attacks (U2R).

The attained Cross-Validation at 2, 5, 10, 30, and 50 folds for each of the four attacks is displayed in Table 15.

Table 15. Achieved Cross-Validation Accuracy at 2, 5, 10, 30, and 50 folds for four attacks

Accuracy at	Attacks			
	Dos	Probe	R2L	U2R
2 folds	0.99662 (+/- 0.00116)	0.99060 (+/- 0.00165)	0.97118 (+/- 0.00143)	0.99519 (+/- 0.00184)
5 folds	0.99709 (+/- 0.00064)	0.99093 (+/- 0.00233)	0.97388 (+/- 0.00624)	0.99714 (+/- 0.00153)
10 folds	0.99738 (+/- 0.00267)	0.99085 (+/- 0.00559)	0.97459 (+/- 0.00910)	0.99652 (+/- 0.00278)
30 folds	0.99726 (+/- 0.00430)	0.99118 (+/- 0.00742)	0.97467 (+/- 0.01644)	0.99693 (+/- 0.00571)
50 folds	0.99703 (+/- 0.00622)	0.99085 (+/- 0.01122)	0.97523 (+/- 0.01795)	0.99662 (+/- 0.00755)

The results obtained in Table 15 indicate consistently high classification accuracy across all attack categories under varying k-fold cross-validation settings, with performance stability influenced by both attack complexity and fold size. DoS and U2R classes exhibit near-saturated accuracy levels (>0.996) across all folds, accompanied by relatively low variance, reflecting strong model generalization and robustness. Probe attacks maintain slightly lower but stable accuracy (~0.991), with a gradual increase in standard deviation as the number of folds increases, suggesting mild sensitivity to data partitioning. In contrast, R2L attacks demonstrate comparatively lower accuracy (ranging from 0.97118 to 0.97523) and a noticeable increase in variance with higher fold values, indicating susceptibility to class imbalance, feature ambiguity, and limited sample representation. Furthermore, increasing the number of folds marginally improves mean accuracy for R2L but at the cost of increased dispersion, highlighting a trade-off between bias reduction and variance inflation.

Overall, the model exhibits robust and stable performance for high-frequency attack classes (DoS, Probe, U2R), while performance for R2L remains comparatively less stable, emphasizing the challenges associated with detecting low-frequency and heterogeneous attack patterns under extensive cross-validation regimes.

IX. ANALYTICAL INSIGHT

The experimental results reveal several important insights into the behavior and limitations of ML classifiers when applied to multi-class ID. A clear pattern observed across all models is the strong performance in detecting majority classes, such as Benign and DoS, contrasted with significantly poor performance in minority classes such as R2L and U2R. This imbalance-driven performance gap highlights a fundamental limitation of conventional ML models, which tend to prioritize overall RE by focusing on dominant classes while neglecting rare but critical attack categories.

Ensemble-based models such as LightGBM, CatBoost, and the Voting Classifier demonstrate relatively strong and consistent performance across majority classes. Their ability to combine multiple decision boundaries or weak learners enables them to capture complex patterns in network traffic data effectively. However, these models also tend to overfit dominant patterns, resulting in high precision but extremely low recall for minority classes. This indicates that while predictions made for rare classes are often correct, the models fail to identify a large proportion of actual instances, making them unreliable for detecting sophisticated and low-frequency attacks.

Neural network-based approaches such as the MLP show moderate generalization capability, achieving balanced performance across common classes but still struggling with minority categories. This suggests that although neural networks can learn complex feature interactions, they require sufficient representative data to perform effectively. Similarly, simpler models such as AdaBoost and SGD demonstrate computational efficiency but fail to capture the intricate patterns required for detecting rare attacks, resulting in poor classification outcomes.

Another important observation is the trade-off between precision and recall across different classifiers. In several cases, high precision is accompanied by extremely low recall, particularly for R2L and U2R classes. This indicates that classifiers adopt a conservative prediction strategy, avoiding false positives at the cost of missing true attack instances. From a cybersecurity perspective, this trade-off is undesirable, as undetected attacks can lead to severe system vulnerabilities. Therefore, achieving a balance between precision and recall is crucial for practical IDSs.

Furthermore, the results underscore the importance of addressing class imbalance in ID tasks. The consistently low detection rates for minority classes across all models indicate that dataset characteristics play a more significant role than model complexity alone. Without appropriate data balancing, feature engineering, or cost-sensitive learning mechanisms, even advanced classifiers fail to perform adequately in critical scenarios.

Overall, the analytical findings emphasize that while ML classifiers are effective for general ID, their reliability in real-world deployments depends on their ability to detect rare and sophisticated attacks. This necessitates the integration of advanced techniques such as resampling strategies, hybrid models, and deep learning approaches to enhance detection performance and ensure robust cybersecurity systems.

X. CONCLUSION & FUTURE SCOPE

The conducted research work evaluated the performance of multiple ML classifiers — LightGBM, Voting Classifier, CatBoost, MLP, AdaBoost, and SGD — on the NSL-KDD dataset, targeting the detection of five traffic classes: Benign, DoS, Probe, R2L, and U2R**.** The models are compared using key evaluation metrics: PR, RE, and F1S**.**

Benign

Table 16 shows the performance comparison of six ML classifiers for the Benign class of the NSL-KDD dataset based on PR, RE, and F1S.

Table 16. Readings of performance metrics for different ML classifiers for the Benign class

Classifier	PR (%)	RE (%)	F1S (%)
LightGBM	72.2	97.1	82.9
Voting Classifier	68.7	97.3	80.6
CatBoost	61.7	97.3	75.6
MLP	70.9	93.5	80.7
AdaBoost	60.9	90.0	72.8

SGD	64	97.4	77.2

Table 16 shows the comparative analysis of the six classifiers for the Benign class shows that LightGBM delivers the most balanced performance, achieving an RE of 97.1%, the highest PR of 72.2%, and the top F1S of 82.9%, indicating strong capability in both correct positive detection and minimizing false alarms. The Voting Classifier and MLP also perform competitively, with RE of 97.3% and 93.5%, respectively, and F1S above 80%, making them reliable choices for balanced ID. While SGD records the highest RE (97.4%), its lower PR (64%) suggests a greater tendency toward FPs. CatBoost achieves high RE (97.3%) but is limited by lower PR (61.7%) and a moderate F1S (75.6%). AdaBoost records the lowest RE (90%), PR (60.9%), and F1S (72.8%), indicating weaker overall performance. Overall, ensemble-based models such as LightGBM and the Voting Classifier emerge as strong contenders for this dataset, with LightGBM offering the most effective balance between PR, RE, and F1S.

DoS

Table 17 shows the performance comparison of six ML classifiers for the DoS attack of the NSL-KDD dataset based on PR, RE, and F1S.

Table 17. Readings of performance metrics for different ML classifiers for DoS attack

Classifier	PR (%)	RE (%)	F1S (%)
LightGBM	96.3	80.5	87.6
Voting Classifier	96.4	78.2	86.3
CatBoost	95.9	83.6	89.3
MLP	90.9	83.6	87.1
AdaBoost	83.5	57.3	67.8
SGD	96.5	77.5	85.9

Table 17 shows the comparative evaluation of the DoS attack on the six ML classifiers, indicating that CatBoost delivers the strongest overall performance, achieving a high PR of 95.9%, a strong RE of 83.6%, and the highest F1S of 89.3%, reflecting its balanced capability in correctly identifying both positive cases and minimizing FPs. MLP matches CatBoost in RE (83.6%) but has lower PR (90.9%) and a slightly reduced F1S (87.1%), suggesting a marginally higher rate of misclassification compared to CatBoost. LightGBM, while having the lowest RE among the top performers (80.6%), records a high PR of 96.3% and an F1S of 87.6%, making it effective where minimizing false positives is critical. The Voting Classifier shows a slightly lower RE (78.2%) compared to LightGBM, but maintains strong PR (96.4%) and a competitive F1S (86.3%). SGD achieves the highest PR of all models (96.5%), but slightly lower RE and F1S compared to LightGBM and the Voting Classifier. AdaBoost underperforms significantly with the lowest RE (57.3%) and F1S (67.8%), indicating limited suitability for this dataset. Overall, CatBoost emerges as the best balanced performer, MLP offers comparable RE, and LightGBM and Voting Classifier excel in PR, making them suitable for scenarios prioritizing low FP rates.

Probe

Table 18 shows the performance comparison of six ML classifiers for the Probe attack of the NSL-KDD dataset based on PR, RE, and F1S.

Table 18. Readings of performance metrics for different ML classifiers for the Probe attack

Classifier	PR (%)	RE (%)	F1S (%)
LightGBM	82.7	63.2	71.7
Voting Classifier	86.2	73.8	79.5
CatBoost	89.6	83.1	86.2

MLP	89.6	83.1	86.2
AdaBoost	82.9	72.5	77.3
SGD	39.9	79.5	53.1

Table 18 shows that CatBoost and MLP share the top position, each achieving the highest PR (89.6%), RE (83.1%), and F1S (86.2%), indicating well-balanced and consistent performance in detecting both positive cases and avoiding excessive FPs. The Voting Classifier follows with an RE of 73.8%, PR of 86.2%, and an F1S of 79.5%, making it a strong performer but slightly less accurate than CatBoost and MLP. AdaBoost records moderate values, with 72.5% RE, 82.9% PR, and a 77.3% F1S, suggesting fair performance but a noticeable drop compared to the top models. LightGBM performs below the mid-tier models, achieving 63.2% RE, although it maintains a reasonably high PR of 82.7%, indicating fewer FPs but more missed detections. SGD demonstrates the most unbalanced performance, with 79.5% RE but an extremely low PR of 39.9% and a poor F1S of 53.1%, suggesting a high rate of FPs despite good RE. Overall, CatBoost and MLP are the most reliable choices for balanced and high-level performance, while the Voting Classifier offers a reasonable trade-off between metrics, and LightGBM may be suitable for scenarios prioritizing PR over RE.

R2L

Table 19 shows the performance comparison of six ML classifiers for the R2L attack of the NSL-KDD dataset based on PR, RE, and F1S.

Table 19. Readings of performance metrics for different ML classifiers for the R2L attack

Classifier	PR (%)	RE (%)	F1S (%)
LightGBM	98.6	8.4	15.4
Voting Classifier	95.7	4.3	8.1
CatBoost	98.6	11.2	20.2
MLP	58.8	13.6	22.1
AdaBoost	---	0	0
SGD	1.0	2.0	4.0

Table 19 shows that all classifiers perform poorly for the R2L class, indicating extreme difficulty in accurate detection. Among them, MLP achieves the highest RE (13.6%) and the top F1S (22.1%), but its PR (58.8%) is considerably lower than the other top-performing models in this group, meaning a higher rate of FPs. CatBoost follows with a high PR (98.6%) and a better RE than LightGBM and Voting Classifier, resulting in a competitive F1S of 20.2%. LightGBM attains a very high PR (98.6%) but only 8.4% RE, leading to a low F1S of 15.4%, which suggests that while its predictions are correct when positive, it misses most actual cases. The Voting Classifier also maintains high PR (95.7%) but records the lowest F1S among the non-zero performers (8.1%) due to extremely low RE (4.3%). Both AdaBoost and SGD perform the worst, with AdaBoost failing to detect any TPs and SGD registering near-zero RE (2%) and an F1S of just 4.0%. Overall, the consistently low RE across models indicates a severe challenge in identifying this class, possibly due to class imbalance or insufficient distinctive features, highlighting the need for targeted techniques such as resampling or specialized algorithms to improve detection.

U2R

Table 20 shows the performance comparison of six ML classifiers for the U2R attack of the NSL-KDD dataset based on PR, RE, and F1S.

Table 20. Readings of performance metrics for different ML classifiers for the U2R attack

Classifier	PR (%)	RE (%)	F1S (%)

LightGBM	43.2	8.0	13.5
Voting Classifier	25.0	0.5	0.9
CatBoost	73.7	7.0	12.9
MLP	---	0	0
AdaBoost	---	0	0
SGD	---	0	0

The results in Table 20 indicate extremely poor performance across all classifiers for the U2R class, suggesting it is highly challenging to detect. LightGBM records an F1S of 13.5%, though its RE is very low (8.0%) and its PR moderate (43.2%), meaning it correctly identifies some TPs but misses the vast majority. CatBoost follows with a notably higher PR (73.7%) than LightGBM, but its low RE (7.0%) keeps the F1S at 12.9%, showing it is very selective but rarely detects positive instances. The Voting Classifier achieves only 0.5% RE, with a PR of 25.0% and an F1S of 0.9%, indicating almost no predictive power for this class. MLP, AdaBoost, and SGD fail, with zero scores across all metrics, implying they could not identify any instances of this class during evaluation. Overall, the consistently low RE across all models reflects a severe imbalance or lack of distinctive features for this category, meaning that without targeted improvements like resampling, cost-sensitive learning, or feature engineering, accurate detection is unlikely.

The underperformance of models such as SGD and AdaBoost can be attributed to their inherent characteristics. SGD, being a linear classifier, struggles to capture complex non-linear relationships present in ID data. Similarly, AdaBoost is sensitive to noisy data and outliers, which are common in imbalanced datasets, leading to degraded performance. These limitations explain their reduced effectiveness in detecting minority attack classes. This leads to a high false positive rate, which in real-world IDSs can result in alert fatigue, overwhelming security analysts with excessive false alarms. To mitigate this issue, cost-sensitive learning approaches can be employed, where higher penalties are assigned to false positives, thereby encouraging the model to improve precision. This highlights the need to balance accuracy with practical usability in cybersecurity environments.

Despite achieving strong performance in the majority of classes, the study highlights a critical limitation in detecting minority attack categories such as R2L and U2R, primarily due to class imbalance and insufficient representative samples. Future research should focus on incorporating data balancing techniques such as SMOTE, cost-sensitive learning, and hybrid deep learning approaches to improve detection rates. Additionally, integrating feature engineering and real-time evaluation frameworks can further enhance IDS performance in practical deployments.

Future work will focus on evaluating the proposed models on more recent and realistic datasets such as CIC-IDS-2017 and UNSW-NB15, which better capture modern network traffic patterns and evolving cyber threats. To enhance detection performance, especially for minority attack classes, future research can explore hybrid modeling approaches. For instance, combining LightGBM, which performs well on the majority of classes, with neural network-based models such as MLP may improve the detection of complex and rare attack patterns. Such ensemble strategies can leverage the strengths of different models to achieve more balanced and robust ID. Due to scope limitations, statistical significance tests were not conducted. Future work will incorporate tests such as paired t-tests to validate performance differences.

XI. COMPUTATIONAL COMPLEXITY ANALYSIS

While detection performance is a primary requirement of an intrusion detection system, computational efficiency is equally important for practical deployment in real-time network environments. The classifiers evaluated in the research paper differ significantly in their training mechanisms, memory requirements, and prediction complexities. Therefore, in addition to classification performance, a qualitative comparison of their computational characteristics is presented to assess their suitability for real-world intrusion detection applications.

The computational complexity of a classifier influences training time, inference speed, resource utilization, and scalability when handling large volumes of network traffic. Furthermore, the use of XGBoost-based feature selection reduced the original feature space to thirteen highly relevant features for each attack category, thereby lowering computational overhead and improving model efficiency. Table 21 presents a comparative assessment of the evaluated classifiers based on their relative computational requirements and practical suitability for real-time intrusion detection systems.

Table 21. Comparative Computational Characteristics of Evaluated Classifiers

Classifier	Training Complexity	Prediction Complexity	Memory Requirement	Relative Training Time	Suitability for Real-Time IDS
LightGBM	Moderate–High	Low	Moderate	Moderate	High
Voting Classifier	High	Moderate	High	High	Moderate
CatBoost	High	Low	Moderate–High	High	High
MLP	High	Low	High	High	Moderate
AdaBoost	Moderate	Low	Moderate	Moderate	Moderate–High
SGD	Low	Very Low	Low	Very Low	High

The comparative analysis indicates that SGD exhibits the lowest computational cost among all evaluated classifiers due to its incremental parameter update mechanism and low memory requirements. AdaBoost also demonstrates moderate computational complexity and can be efficiently trained on medium-sized datasets. Ensemble-based approaches such as LightGBM and CatBoost require comparatively higher computational resources during training because multiple decision trees are generated iteratively; however, they provide efficient prediction performance after model training. The Voting Classifier incurs additional computational overhead as it combines the outputs of multiple constituent models. MLP exhibits relatively high computational complexity because of iterative weight optimization through backpropagation. Despite these differences, the application of XGBoost-based feature selection reduces the dimensionality of the dataset to thirteen features per attack category, thereby lowering computational overhead and improving the feasibility of deploying the proposed framework in practical ID environments.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

The data will be available on request to the corresponding author.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

AJ, MS, VKB: Writing, Investigation, Data collection

AJ, JK, GJ, SH: Software, Validation, Conceptualization

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